



Cisco Catalyst Center Assurance Tools Overview

Turning insights into action: Assurance tools that drive trust and performance

This eBook serves as a quick-reference guide for diagnosing and resolving issues through Catalyst Center's assurance dashboards and toolsets. It highlights the most effective workflows, provides step-by-step troubleshooting support, and surfaces proven practices for keeping networks healthy and resilient.

Select one of the tabs or turn the page to get started.

Introduction

**Troubleshooting
Tools**

**Key
Takeaways**

Why Structured Troubleshooting Matters

Modern enterprise networks generate vast amounts of telemetry data, making traditional reactive troubleshooting approaches inefficient and time-consuming. Catalyst Center's assurance tools enable every point on the network to become a sensor, sending continuous streaming telemetry on application performance and user connectivity in real time. This, coupled with automatic path-trace visibility and guided remediation, means network issues are resolved in minutes—before they become problems.

Challenges of Modern Network Troubleshooting:



Complexity Scale

Today's networks span multiple domains (wireless, wired, SD-WAN) with thousands of devices

User Expectations

Zero-tolerance for downtime and performance degradation



Data Overload

Network devices generate massive amounts of logs and metrics

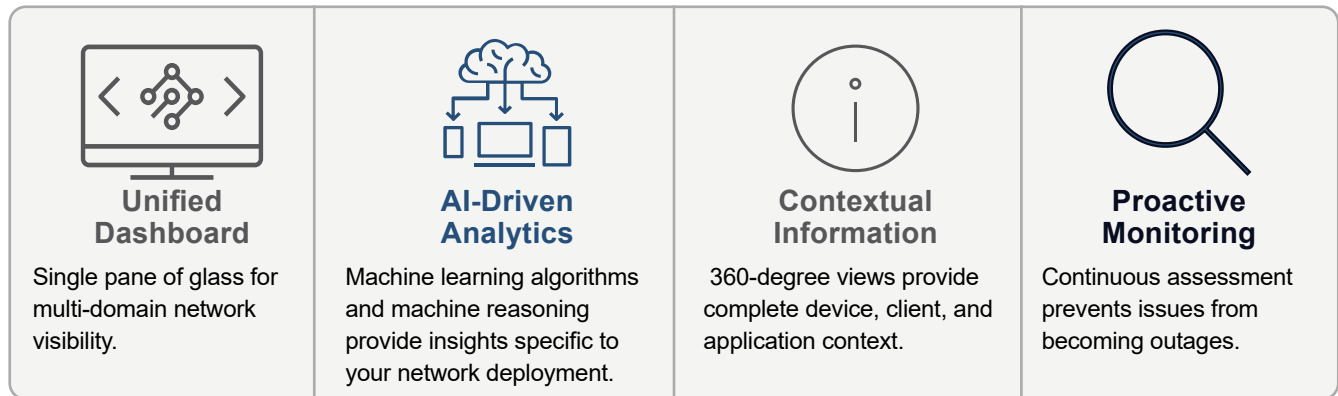
Skill Gaps

Traditional CLI-based troubleshooting doesn't scale with network complexity



Catalyst Center's Solution Approach

To address these challenges, Catalyst Center brings together advanced assurance tools designed to simplify complex networks and reduce troubleshooting time. Its approach leverages AI, contextual insights, and continuous monitoring to provide administrators with the clarity they need to act quickly.



By combining these capabilities, Catalyst Center transforms raw telemetry data into actionable intelligence. The platform doesn't just provide visibility, it guides administrators through the process of identifying, analyzing, and resolving issues before they impact users. This structured approach to troubleshooting ensures that insights are consistent, scalable, and aligned with the growing complexity of modern networks.

Benefits of Structured Troubleshooting

-  **Faster Resolution**
Event Analytics provides network administrators insights into the events that are most relevant for them.
-  **Improved Accuracy**
Event Analytics provides network administrators insights into the events that are most relevant for them.
-  **Knowledge Transfer**
Event Analytics provides network administrators insights into the events that are most relevant for them.
-  **Preventative Action**
Event Analytics provides network administrators insights into the events that are most relevant for them.

Core Tools for Troubleshooting

Catalyst Center's solution approach comes to life through a set of purpose-built troubleshooting tools. Each tool is designed to give administrators clear visibility into different aspects of the network, whether it's a single device, an individual client, or an entire application flow. By combining these perspectives, teams can pinpoint issues faster, validate root causes, and ensure that performance and reliability stay aligned with business needs.

Explore the Tools

Select each tool to explore key features and learn how it helps resolve issues.

Device 360

Check device health, performance issues, and configuration drift

Application Health Monitoring

Diagnose application slowdowns, bandwidth, or QoS concerns

Event Analytics

Identify network-wide issues, recurring problems, and root causes

Path Trace

Verify routing, policy flows, and end-to-end connectivity gaps

Intelligent Capture and Over-the-Air Sniffer

Investigate packet details and troubleshoot wireless channels

Device 360

Device 360 is a comprehensive device health and performance dashboard providing real-time and historical insights into individual network devices. Network health scores and KPI metrics do not include network devices that are in maintenance mode.

**View the
Workflow**

[Click Here](#)

Key Features	Description	When to Use Device 360
Health Dashboard	Overall device health score with contributing factors	Device appears offline or unreachable
Performance Metrics	CPU, memory, temperature, and interface utilization	Performance complaints from users connected to specific switches
Connectivity Status	Port status, neighbor relationships, and link health	High CPU or memory utilization alerts
Configuration Management	Current vs. intended configuration comparison	Configuration drift investigations
Event Timeline	Chronological view of device events and state changes	Interface errors or high utilization warnings
Maintenance Mode Integration	Devices in maintenance mode show special banners and are excluded from health calculations	Investigating devices with recurring issues (use device tagging for history)

Critical Client States

Onboarding Experience	Association, authentication, DHCP, DNS resolution
Connected Experience	Signal strength, data rates, roaming behavior
Application Experience	Response times, throughput, packet loss per application

Key Metrics to Monitor

- **Health Score:** Composite score (0-100 indicating overall device wellness)
- **Reachability:** ICMP and SNMP connectivity status
- **Interface Statistics:** Error rates, utilization percentages, duplex mismatches
- **Environmental:** Temperature readings, power supply status, fan speeds
- **Performance Threats:** Historical graphs showing usage patterns over time

Notice of Limitations in Version 2.3.7

This version does not support remote LAN (RLAN) or RLAN clients, and it also does not support workgroup bridge (WGB) clients—though WGB itself will still appear as a wireless client.

Application Health Monitoring

The application-centric view highlights how network performance impacts specific business applications, leveraging Application Visibility and Control (AVC) and Application Telemetry to continuously monitor and optimize application performance.

View the Workflow
Click Here

Key Features	Description	When to Use Application Health Monitoring
Application Health Scoring	Real-time health metrics for monitored applications	Application performance complaints affecting multiple users
Traffic Analysis	Volume, patterns, and quality metrics per application using NetFlow and streaming telemetry	Bandwidth planning and capacity analysis
User Impact Assessment	Number of affected users and severity levels	QoS policy effectiveness validation

Premium Content

You've reached the end of the preview content.

The complete troubleshooting guide includes additional detail pages and workflows for:

- Application Health Monitoring
- Event Analytics
- Path Trace
- Intelligent Capture and Over-the-Air Sniffer

To view the complete Folio contact your account manager

Event Analytics

Event Analytics provide AI-powered centralized event correlation and analysis platform that processes network events, logs, and alarms across all managed devices. In version 2.3.7, event data retention is increased to 60 days, and cloud connectivity is required.

**View the
Workflow**
[Click Here](#)

Key Features	Description	When to Use Event Analytics
AI-Powered Event Correlation	Links related events across different devices and timeframes using machine learning	Network-wide issues affecting multiple devices or sites
Root Cause Analysis	Automated identification of underlying issues	Pattern recognition for recurring problems
60-Day Data Retention	Extended event history for better trend analysis (new in 2.3.7)	Compliance reporting and audit preparation
Custom Dashboards	Configurable views for different operational teams	Proactive monitoring and alerting setup
Alert Management	Severity-based filtering and escalation workflows	Post-incident analysis and documentation

Event Categories

- **Critical:** Service-affecting events requiring immediate attention
- **Major:** Significant issues that may impact service qualityRequires Advantage license for full functionality
- **Minor:** Notable events that should be monitored but don't require immediate action
- **Warning:** Potential issues that may escalate if not addressed
- **Informational:** Normal operational events for audit trails

Path Trace

Path Trace provides a detailed, end-to-end visualization of the network path between two nodes, enabling administrators to analyze performance, identify bottlenecks, and validate connectivity across the infrastructure.

**View the
Workflow**

[Click Here](#)

Key Features	Description	When to Use Path Trace
End-to-End Visualization	Complete network path from source to destination	Routing issues between network segments
Hop-by-Hop Analysis	Shows devices along the path with their IP addresses	Policy verification for traffic flows
Protocol Support	Works with TCP or UDP traffic	Connectivity troubleshooting between hosts or Layer 3 interfaces

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Intelligent Capture and Over-the-Air Sniffer

Advanced packet-level analysis tools providing detailed packet inspection capabilities. Version 2.3.7 introduces the Over-the-Air Sniffer, which expands on Intelligent Capture by listening to more than just packets between a single client and access point.

View the Workflow

Click Here

Intelligent Capture

Key Features	Description
Targeted Packet Capture	Selective capture based on specific criteria
AI-Powered Analysis	Machine learning-based packet analysis and anomaly detection
Performance Insights	Deep-dive into application behavior and network interactions
Export Capabilities	Integration with external analysis tools

When to Use Intelligent Capture

Best suited for analyzing application behavior in detail, conducting security investigations, and performing in-depth troubleshooting where precise packet-level visibility and long-term accuracy are required to resolve complex issues.

Now in 2.3.7 Over-the-Air Sniffer

Key Features	Description
Radio-Wide Monitoring	Selective capture based on specific criteria
Multi-Band Support	Machine learning-based packet analysis and anomaly detection
Floor Map Integration	Deep-dive into application behavior and network interactions
Wireless Troubleshooting	Integration with external analysis tools

When to Use Over-the-Air Sniffer

Ideal for diagnosing wireless connectivity issues, investigating poor performance, and addressing channel-wide problems that impact multiple clients or nearby access points across different bands and frequencies.

Key Takeaways, Resources, and Support

Cisco Catalyst Center gives JPMC teams the tools to manage complex networks with greater clarity and control. From real-time visibility to automated compliance and smart integrations, it supports efficient, consistent operations across regions. With standardized templates, AI-driven insights, and connected workflows, teams can respond faster and scale with confidence.

A strong knowledge network is just as valuable as a strong infrastructure. Whether you need documentation, quick templates, or a teammate to review your next deployment plan, this section gives you a direct line to the resources that matter.

Note to SMEs:
This is an example and will differ based on your internal resources. We'll need a list of additional tools and resources that you would like to include.

Tools and Templates

- 15-Minute Troubleshooting Checklist (fillable PDF or form)
- Golden CLI Templates (link to repo or OneDrive)
- Troubleshooting Workbook (interactive PDF or web form)
- Weekly Health Review Template (PPT or Google Slides)

Further Reading and Reference

- Catalyst Center Guides (User, Assurance, Platform, Rogue Mgmt, Stealthwatch)
- Cisco Learning Network videos (e.g., Device 360, MRE in Action)
- Support Community and internal playbooks

**Stay aligned, stay
proactive, and let
Catalyst Center do
the heavy lifting.**

